# ALDO VILLAMIL

AI ARCHITEK · AUTONOMOUS AGENT SYSTEMS · MCP · BLOCKCHAIN · CRM PLATFORMS · CO-FOUNDER

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| | | | | | |
|---|---|---|---|---|---|
| 3 | 600+ | 4 | 3 | 3 | 20+ |
| LIVE PRODUCTS | AGENT SKILLS | OPEN-SOURCE | AMD CERTS | CHAINS SHIPPED | YEARS CS/OPS |

| | | | |
|---|---|---|---|
| **AGENTS** | Hermes fleet, sub-agent swarm, council of models, MCP plugins | **BACKEND** | Rust axum, Go, PHP Laravel, TypeScript Next.js, Python FastAPI |
| **ON-CHAIN** | Solana, Base L2, Ethereum, Solidity, x402 and c402 HTTP 402 | **DATA** | PostgreSQL, MySQL, MongoDB, SQLite, Redis, Prisma, Neon |
| **CRM** | Multi-tenant Laravel, GoHighLevel pipeline mapping, HMAC webhooks | **OPS** | DigitalOcean, Vercel, GitHub Actions, Docker, Kubernetes |

## Summary

**AI Architek designing and running autonomous multi-agent systems in production.** Not one harness: a fleet of them — Hermes, a DeepSeek harness, and focused worker harnesses, each a full runtime with its own memory, sessions and scheduled work — routing tasks to specialist sub-agents (OpenCode / Cline / Codex). A council of models, each LLM an independent policy aggregated by a reward model, returns one high-confidence answer; persistent cross-session memory keeps design intent alive. 600+ versioned skills load on demand; when the agents lack a capability, I build it: a new agent, a sub-agent, a custom skill, or an MCP plugin. **Shipped end-to-end:** an AI real-estate platform with Colombian legal compliance (**Pequi, live**), an on-chain **x402** payment facilitator on **Base mainnet** listed in the **Coinbase ecosystem**, Colombia's first public real-estate API monetized with the **c402 HTTP-402 protocol**, a multi-tenant CRM platform with client portals, and a **Solana** execution stack proven on mainnet with real money. **Go, Rust, PHP, TypeScript and Python** in production, plus CRM middleware on axum (HMAC webhooks, SSE fan-out, rate limiting). Published open-source tooling in three languages: an MCP contract probe (Go), a webhook signature guard (PHP, 11/11 self-tests), and a constant-time HMAC verifier (Rust). MCP servers on the latest 2026-07-28 stateless Streamable HTTP spec, verified against a published tool contract so a silent drop cannot reach production. Bilingual EN/ES, US time zones, owner from architecture to deployment.

## Verified In Production

// measured, not estimated · every number below came from a running system

**201ms**
Solana slot time measured on the live box

**3.08%**
Real round-trip cost, fees plus priority

**98.3%**
Exit retention proven on a simulated swap path

**8453**
Base chain id for live USDC settlement

**2,342**
Lines of axum serving the client portal

**600+**
Versioned skills loaded on demand by the fleet

**276**
Barrios served by the live API and platform

**11/11**
Self-tests passing in the published guard

## AI Architecture & Agent Systems

SWARM // 24/7

### Agent Swarm

// orchestrator decides · routes · verifies

Specialist sub-agents build in parallel — one task per agent, isolated workspaces. No self-report trusted until verified. Runs 24/7 in production on DigitalOcean with an always-on security lane.

COUNCIL // RELIABILITY

### Council of Models

// reliability at the answer level

High-stakes questions routed to specialist LLMs as independent policies. Reward-model aggregation returns the single highest-confidence answer.

PIPELINE // SPEC-FIRST

### Verification & Security Lane

// spec-first pipeline

3-stage gate before merge: lint → typecheck → integration. Secret-scanning across repos/logs, credential hygiene, spend monitoring.

## Blockchain & Payments Engineering

// solana mainnet · base layer 2 · ethereum · agent payments

SOLANA // MAINNET

### On-Chain Execution Stack

// verified executor · real money

A verified executor proven with real money on mainnet: $1 round trips landed flat, fresh versus existing token accounts costed, and Jupiter and Meteora DLMM swap paths simulated before signing. Latency measured against a live competitor feed rather than assumed, and a first measurement that looked 11× better was corrected to the real 1.3×.

201 ms/slot · 3.08% round-trip cost · 98.3% exit retention

BASE L2 // ETHEREUM

### x402 Payment Facilitator

// http 402 · on-chain settlement

HTTP 402 payment negotiation with on-chain USDC settlement and on-chain verification, agent to agent. Solidity contracts, ERC-1155 token gating, Chainlink VRF randomness. Listed in the Coinbase ecosystem, with full test coverage.

chainId 8453 · eip155:8453 · Base Sepolia testnet

C402 // FIAT RAILS

### HTTP 402 for Colombia

// pay-per-request · no wallet

Colombia-first HTTP 402 protocol: agents and applications pay per request in COP over the rails Colombians already use, with zero-crypto settlement. Challenges carry a nonce and TTL and are replay-guarded. The fiat counterpart of x402, authored as a public MIT spec.

spec v2.1 · Wompi · PSE · Nequi · Daviplata

## CRM & Client Platform Engineering

// multi-tenant · gohighlevel · webhooks · ai ops

LARAVEL + RUST // MULTI-TENANT

### Agency CRM & Client Portals

// one client, one pipeline

A Laravel 10 agency platform with tenant-isolated client records and per-client GoHighLevel pipeline mapping, so every client syncs to its own marketing pipeline instead of a shared one. A 2,342-line axum portal serves the client dashboard, deployed and operated on DigitalOcean with daily sync jobs, and every client mapping is protected by a unique client identifier so a webhook can never cross tenants.

strategy-pattern payload builders · unique client_id mapping · GHL 150+ markets

RUST // AXUM + SQLITE

### CRM Middleware & AI Operations

// fail-closed · receipts written

Axum middleware over SQLite that serves the embedded PWA, verifies HMAC-signed webhooks with replay protection, and rate-limits by tenant. An AI backfill runs nightly behind a fail-closed gate that writes a receipt for every decision, and a read-only browser agent audits the platform weekly. Deploys are gated on an artifact marker that only the new binary emits, so a stale build can never be published. A published PHP guard covers the same HMAC and RSA-SHA256 verification path with zero dependencies.

fail-closed · receipts logged · read-only by design · 11/11 self-tests

## Projects — S.D.A.E. Ecosystem

// Sustainable · Decentralized · Autonomous · Ecosystem

### 01 Pequi LIVE

`Next.js 16` `React 19` `Clerk` `Wompi` `Prisma` `DeepSeek V4` `GRPO roadmap`

AI real-estate platform for Colombia. Ley 820-compliant contracts, payments, 15 AI tools, multi-agent chat. Live in Ibagué + Bogotá (276 barrios), c402 monetization.

VISIT →

### 02 MCPVOTS LIVE

`Solidity` `Base` `Chainlink VRF` `ERC-1155` `USDC` `Hardhat`

"Stripe for AI agents": on-chain x402 payment facilitator — autonomous agent-to-agent payments on Base mainnet. Coinbase-ecosystem listed; full test coverage, on-chain verification.

VISIT →

### 03 MCPVotsAGI

`Python` `AsyncIO` `WebSockets` `Redis` `PyPI`

Multi-model consensus SDK — specialist LLM outputs aggregated by reward model into one high-confidence answer. A2A protocol gateway, Redis-backed persistent memory. **pip install mcpvotsagi**

VISIT →

// PROTOCOL — 04

### xpequi-api + c402 Protocol

`REST API` `Open Finance` `TS SDK` `Python SDK` `MCP Server` `IPFS`

Colombia's first public real-estate data API — **18 endpoints, 276 barrios**, with a stateless MCP server on the 2026-07-28 spec. c402 adds Colombia-first HTTP 402 fiat rails (Wompi · PSE · Nequi · Daviplata) with zero-crypto settlement — agents and apps pay per request through local payment rails, no wallet required.

VISIT XPEQUI.XYZ →

## Tech Stack

LANGUAGES

`TypeScript` `Python` `Go` `PHP`
`Rust` `Solidity` `SQL`

AI / AGENTS

`DeepSeek V4` `Hermes Agent`
`OpenCode` `Cline` `Codex`
`OpenRouter` `Groq`
`Council of Models`

MCP

`2026-07-28 Spec`
`Stateless Servers`
`Streamable HTTP` `MCP Server`
`Agentic Wallet`

BLOCKCHAIN

`Solana` `Base` `Ethereum`
`Solidity` `Chainlink VRF`
`ERC-1155` `Jupiter` `Hardhat`

BACKEND / DB

`PostgreSQL` `MySQL` `MongoDB`
`SQLite` `Prisma` `Neon` `Redis`
`Go` `Rust (axum)` `CRM middleware`

FRONTEND

`Next.js` `React`
`HTMX (server-driven)`
`Tailwind CSS` `Three.js`

AUTOMATION

`n8n` `DeerFlow` `VoltAgent`
`Cron Jobs` `Message Queues`
`GHL 150+ markets`

DEVOPS

`GitHub Actions` `Vercel`
`DigitalOcean` `Docker`
`Kubernetes (AMD)` `Sentry` `Nginx`

ML / RL

`PPO` `GRPO` `RLVR`
`Reward-Model Aggregation`

## Work Experience

### Co-Founder & Full-Stack Developer — Village of Thousands

2021 – Present · Remote

- Headless e-commerce: React + Shopify on Vercel, DigitalOcean backend (99.9% uptime).
- Created brand identity — graffiti-inspired design language; NYFW-featured.

### Digital Marketing Manager — PolloRiko.io

2018 – Present · Houston, TX (Remote)

- Full digital presence for US restaurant chain: web, SEO, social, AI-generated content.

### Customer Service — United Health Care (Teleperformance)

2020 – 2021 · Ibagué, Colombia

- EN-language support for US members; high CSAT on US time zones.

### Banquet Captain — Hilton Bonnet Creek

2017 – 2018 · Orlando, FL

- Led service teams for large-scale corporate events; luxury service standards.

## Education & Certifications

### Computer Programming and Analysis

**Valencia Community College**
Orlando, FL · 2004 – 2008

### AMD AI Academy — Certified

**3 AMD Certifications**

- AI on AMD Instinct GPUs
- Hugging Face on AMD
- Kubernetes on AMD Instinct

*Continuous self-directed learning verified through shipped production products.*